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RÉSUMÉ – L'analyse sémique est une méthode linguistique par laquelle le sens dénotatif et connotatif des termes est décomposé en sèmes. Dans ce travail, nous présentons une ressource multilingue axée sur l'analyse sémique de la terminologie médicale, visant à : 1) fournir les ensembles de sèmes dénotatifs et connotatifs des termes, 2) donner des informations sur la connotation des termes, et 3) fournir un inventaire des sèmes. La ressource s'adresse aux patients, médecins et professionnels de la langue.

MOTS-CLÉS – dénotation, connotation, analyse sémique, terminologie médicale, ressource terminologique multilingue

BONATO (Vanessa), DI NUNZIO (Giorgio Maria), VEZZANI (Federica), « L'interaction entre dénotation et connotation : application de l'analyse sémique à la terminologie médicale »

ABSTRACT – Semic analysis is a linguistic method through which the denotative and the connotative sense of terms is decomposed into semes. In this work, we present a multilingual resource that focuses on semic analysis for medical terminology, designed to: 1) provide the sets of denotative and connotative semes of terms, 2) convey information on the connotation of terms, and 3) supply an inventory of semes. This resource is intended for patients, physicians and language professionals.

KEYWORDS – denotation, connotation, semic analysis, medical terminology, multilingual terminology resource

THE INTERPLAY BETWEEN DENOTATION AND CONNOTATION: APPLYING SEMIC ANALYSIS TO MEDICAL TERMINOLOGY

Introduction

Investigating the semantic dimension of terminology involves the necessity to examine the denotation and the connotation attributed to terms. Denotation constitutes the neutral meaning that is conventionally associated to a lexical unit (Bloomfield 1933). By neutral we mean that denotation is objective (Vezzani 2023) and, therefore, devoid of any personal deviations (Bloomfield 1933). Connotation, on the other hand, is the set of “supplementary values” acquired by a lexical unit due to cultural, linguistic, or social influences (Bloomfield 1933).

Insofar as medical terminology is concerned, denotation plays a fundamental role. In particular, medical language is supposed to be characterized by emotional neutrality and text objectivity, therefore greater relevance is given to denotation over connotation (Serianni 2003). However, connotation also significantly impacts medical terms. As a matter of fact, as stated by Balliu (2005: 75):

*[u]n texte médical ne peut faire l'économie du fait qu'en fin de compte, il concerne toujours l'homme, dans ce qu'il a de plus humain et de plus fragile, la vie.*¹

¹ A medical text cannot ignore the fact that, in the end, it always concerns human beings and what is most human and fragile of human beings: life.

For example, a negative connotation is attributed to the medical terms *hypochondriasis*, *somatization* and *hysterical neurosis* (Dury 2012). From a terminological perspective, the negative connotation acquired by medical terminology (Dury 2012, 2013; Vezzani 2023; Vezzani and Costa 2024) can eventually lead to the occurrence of term variation (Dury 2022; Freixa 2022). Beyond that, “[p]atients may perceive medical language as stigmatizing, regardless of the physician’s intent” (Losorelli, Su-Velez, and Balakrishnan 2023). More particularly, the use by physicians of terms that are “widely accepted and understood to those conversant in the vocabulary of medicine, but that have nonmedical definitions with negative associations” can “be a barrier to patient understanding of the disease process” (*ibid.*). This is indicative of the fact that the connotation of medical terms exerts a major influence also in healthcare settings.

The consideration of the different role and impact of denotation and connotation of medical terminology highlights the necessity to respectively examine both phenomena. Such an investigation can support the acquisition of a sound comprehensive knowledge of the semantic dimension of medical terms. To this end, semic analysis (Rastier 1987) constitutes a valuable approach. Semic analysis is a linguistic method which can be used to represent the denotative and the connotative sense of lexical units by decomposing it into semes. Specifically, semes are considered as atoms of meaning that cannot undergo further factorization. In particular, the sense of terms can be expressed by denotative semes and connotative semes. The respective investigation of these two categories of semes, however, requires to systematically distinguish between the set of denotative and connotative semes of medical terms.

In this work, we present the implementation of a novel multilingual and multipurpose terminology resource that focuses on semic analysis for medical terminology. The resource addresses different categories of users: patients, physicians, and language professionals. The main goal is to provide the user with a tool that is designed to: 1) distinctively provide the sets of denotative

and connotative senses of medical terms, 2) convey information concerning the potential positive or negative connotation attached to terms, and 3) supply an inventory of senses in the form of dedicated medical terminological entries. The specific purposes of the tool are: 1) to serve as an aid for the acquisition of knowledge about denotation and connotation of medical terms by patients, 2) to support language professionals in the interlinguistic analysis of medical terminology at the semic level, and ultimately 3) to facilitate patient-physician communication.

The remainder of the paper is the following. In Section 1, we present background studies on denotation and connotation of medical terminology. We also consider contributions that concern semic analysis of medical terms and term polarity. In Section 2, we focus on the theoretical approach that underlies the design of the resource. In Section 3, we describe the resource. In Section 4, we present a case study to illustrate the implementation and functionality of the resource. In particular, a demo of the resource is contextually presented. In Section 5, we provide conclusions and directions for future work.

1. State-of-the-art

In this section, we present an overview of background studies concerning denotation and connotation, focusing on their relation to medical terminology (1.1). Furthermore, we consider state-of-the-art works on semic analysis of medical terms (1.2) and term polarity (1.3).

1.1. *Denotation and connotation in medical terminology*

Medical terminology lies in the domain of special languages, which covers multiple languages purposefully used to communicate and exchange information within specialized fields of knowledge (Cortelazzo 1990). As far as terminology belonging to special languages is concerned, “terms [are supposed to] have a purely denotative function” (Gotti 2008: 35). This feature differentiates

terms from general-language lexical units, that are “often richly connotated” (*ibid.*). The primacy of denotation over connotation should characterize terminology belonging to the healthcare domain, ensuring objectivity (Morales Harley 2014) and emotional neutrality (Serianni 2003). For example, in medical specialized language, the denotation of somatic terms is supposed to prevail over connotation to guarantee absence of ambiguity (Vezzani 2023). Connotation, however, is also present in specialized texts (Gotti 2008). For instance, it can be detected in specialized texts that pertain to the oncology domain (Vezzani 2023).

Connotation, in particular, is a linguistic phenomenon that exerts a significant impact on medical terminology. From a terminological perspective, indeed, the connotative sense plays a considerable role in the diachronic evolution of medical terms. As previously mentioned, the negative connotative value acquired by medical terminology (Dury 2012, 2013; Vezzani 2023; Vezzani and Costa 2024) leads to term variation (Dury 2022; Freixa 2022). Specifically, the pejorative connotative charge attributed to terms that belong to the healthcare domain lies at the foundation of the “neological need” perceived by specialists (Dury 2012, 2013). This means that, at times, medical experts need to create new terms for existing medical concepts to replace terminology that acquired negative connotations. As an example, the negatively connotated English term *vegetative state* was replaced by the terms *unresponsive wakefulness syndrome* and *minimally conscious state*, which specialists regard as more scientifically precise and unbiased (Dury 2013). Another example of term variation due to connotation is presented in the work by León-Araúz (2017: 216):

[...] it was only after the discovery that *Down syndrome* was caused by the presence of an extra chromosome that the term variant *trisomy 21* emerged, which focuses on the causal dimension of the concept. Until then, the variants focused on the discoverer of the disease (John Langdon Down 1828–1896) and the physical attributes of the patients (*mongolism*). This term was eventually suppressed because of its obviously negative connotations.

The stigmatizing connotation attached to medical terms can also negatively affect the perception of medical conditions, as in the case of the term *schizophrenia* (Lasalvia, Penta, et al. 2015; Lasalvia, Vita, et al. 2021). As a matter of fact, “[m]ental health users and professionals around the world have started calling for a change of the name, seeing it as stigmatizing and harmful” (Lasalvia, Penta, et al. 2015: 276). Beyond that, from a multilingual standpoint, the consideration of the connotative sense attributed to medical terms becomes an essential aspect to address during the translation process (Vezzani 2023). The author, indeed, highlights that connotation is fundamentally involved in the achievement of translation equivalence and influences translation choices. As indicated in Vezzani (2023), the Italian term *mammella* should not be translated into French as *mamelle*, due to the pejorative connotative charge diachronically acquired by *mamelle*.² The correct translation equivalent for *mammella* is the French term *sein*, which is not negatively connotated. As previously mentioned, connotation is also determined by cultural influences (Bloomfield 1933). For what concerns communication, in particular, the cultural dimension is an essential element to consider in the medical terminological domain (Bobokalonov 2023: 159):

[c]ultural sensitivity in medical terminology is imperative for ensuring effective communication in healthcare settings [...]. The interpretation of health, illness, and medical concepts is inherently influenced by cultural factors, making cultural sensitivity an integral aspect of linguistic precision in medical language.

1.2. *Semic analysis of medical terminology*

Semic analysis makes it possible to factorize the sense of terms into denotative semes and connotative semes that comprehensively constitute a sememe (Rastier 1987). By adopting the terminology

² For a detailed description of the extension of the connotation of the term *mamelle*, see Vezzani (2023).

used by François Rastier, denotation and connotation are respectively represented in the form of inherent semes and afferent semes.

The linguistic technique of semic analysis, in particular, is adopted by the software SEMY, that provides a “dictionnaire de traits infralexicaux (ou sèmes) [...] réalisé à partir d’une extraction des mots des définition d’un dictionnaire de langue informatisé, le *Trésor de la Langue Française informatisé* (Dendien & Pierrel 2003)” (Grzesitchak, Jacquey, and Valette 2007: 229) and “aims to give a semi-automatic semantic profile of texts or linguistic entities” (Baider 2014). The objective that underlies the approach adopted by the authors is to “évaluer le potentiel de l’analyse sémique en matière d’identification du contenu sémantique d’un texte” (Grzesitchak, Jacquey, and Valette 2007: 227).

Semic analysis has been used to investigate medical terminology in multiple studies, concerning for instance terms related to physical defects (Stan 2014), the pharmaceutical field (Pirlogea 2023) as well as causative verbs (Staicu 2021). Moreover, semic analysis is provided in the multilingual terminology database TriMED (Vezzani and Di Nunzio 2020), that inspired the development of the resource presented in the current work. In particular, TriMED focuses on medical terms and is purposefully created to support multiple categories of users “with problems related to the opacity of the medical vocabulary for different aspects and different levels of communication: physicians, patients and scientific translators” (Vezzani, Di Nunzio, and Henrot 2018: 4368). Semic analysis of medical terms is also the focus of a study carried out by Bonato, Di Nunzio, and Vezzani (2021). The objective of the work is to systematize the performance of semic analysis through the creation and application of a methodology based on two criteria. The contribution also includes an experimental study aimed at investigating the existence of an interrelation between semic analysis and word embeddings. In a more recent work, Bonato, Di Nunzio, and Vezzani (2024) presented “an extension, development, and integration of the methodology for a systematic semic analysis of medical terminology.” In this paper, the distinction between

denotative and connotative senses is used to study polysemy and polyreferentiality of medical terms. Furthermore, the connotative senses associated to polysemous terms are examined to evaluate the positive or negative connotation associated to the analyzed terms.

1.3. *Term polarity*

Terminology can contextually acquire connotation, which is a phenomenon that exclusively pertains to the linguistic dimension of terms. The connotative charge that is attributed to terminology can be essentially considered as the polarity of terms. According to Strapparava and Mihalcea (2007: 70):

[a]ll words can potentially convey affective meaning. Every word, even those apparently neutral, can evoke pleasant or painful experiences due to their semantic relation with emotional concepts or categories.

The identification of polarity constitutes “[t]he most popular task of sentiment analysis” (Martinis, Zucco, and Cannataro 2022: 1), the latter of which is a technique that is also applied to the healthcare setting (Kerstin Denecke and Deng 2015; Satapathy, Cambria, and Hussain 2017; Abualigah et al. 2020; Zunic, Corcoran, and Spasic 2020; Kerstin Denecke 2023b). Specifically, as defined by Kerstin Denecke (2023b: 3), “[m]edical sentiment analysis refers to the identification and analysis of sentiments or emotions expressed in free-textual documents with a scope on healthcare and medicine.” For example, its application can be useful “to collect statistical information on the outcome of treatments” (Kerstin Denecke and Deng 2015: 25) or on the part of physicians “to identify and summarize the health status of a patient [...]” (*ibid.*). The analyzed documents can be, among others, clinical narratives (Deng, Stoeck, and K. Denecke 2014; Kerstin Denecke and Reichenpfader 2023), discharge notes from hospitals (McCoy et al. 2015), forums dedicated to the medical domain (Ali et al. 2013; Bobicev and Sokolova 2018) and texts found in social media (Yadav et al. 2018; Sharma et al. 2020; Babu and Kanaga 2021). In

the context of medical sentiment analysis, polarity detection plays a relevant role. In particular, to give an example, “[i]dentifying emotional polarization in a medical report is important in screening, acquiring and synthesizing knowledge of physicians before making a clinical decision” (T. D. Nguyen, L. D.-P. Nguyen, and Cao 2017: 49).

Term polarity has been specifically considered in multiple studies related to medical terminology. For instance, dedicated resources providing lists of medical terms annotated for their polarity have been developed, such as medical opinion lexicons (Goeuriot et al. 2012; Asghar, Khan, et al. 2014) and sentiment lexicons (Asghar, Ahmad, et al. 2016; Kerstin Denecke 2023a).

2. Theoretical background

The theoretical approach that we adopt in this paper assumes that the discipline of terminology possesses a twofold dimension: conceptual and linguistic (Costa 2013; Santos and Costa 2015). These complementary dimensions are interrelated, but respectively focus on different entities. Indeed, the conceptual dimension is specifically concerned with concepts. The linguistic dimension, instead, deals with terms as linguistic verbalizations of concepts. We adhere to the definition of *term* provided by the ISO 1087 (2019), according to which the term is a “*designation* that represents a *general concept* by linguistic means” which “may be partly of wholly verbal.”

For what concerns the conceptual dimension, we consider the *concept* as a “unit of knowledge created by a unique combination of *characteristics*” (ISO 1087 2019). The characteristics are abstractions of the properties of the object, the latter being defined as “anything perceivable or conceivable” (*ibid.*). In particular, the characteristics can be extracted from intensional definitions. In line with the ISO 1087 (2019), we consider an intensional definition as a “*definition* that conveys the *intension* of a *concept* by stating the immediate *generic concept* and the *delimiting characteristic(s)*.”

Focusing on the linguistic dimension, instead, each term possesses *semes* (Vezzani, Piccini, et al. 2023). In this study, we specifically adhere to one of the main distinctions concerning *semes*, which is the one between denotative *semes* and connotative *semes* (Depecker 2002).

Following the approach presented by Vezzani, Piccini, et al. (2023), the characteristics at the conceptual level match the set of denotative *semes* possessed by the term at the linguistic level. Connotative *semes*, instead, are attached to terms used in specific discourses and consequently do not correspond to the characteristics that make up the concept. When connotative *semes* are anchored to a term in discourse, the term acquires multiple senses. According to Vezzani, Piccini, et al. (2023), polysemy in terminology concerns the multiplicity of senses acquired by a term, that does not entail the multiplicity of concepts. Polyreferentiality, instead, is a phenomenon that concerns the conceptual dimension of terms. It occurs when different terms that uniquely have in common the same lexical form designate different concepts. In other words, in polyreferentiality, multiple terms share the same sequence of characters and, as a consequence, they are identical from a lexical viewpoint. However, these terms differ one from the others, due to the fact that each term designates a different concept.

In this work, we adopt the approach proposed by Bonato, Di Nunzio, and Vezzani (2024), which consists of the application of *semic analysis* to study polysemy and polyreferentiality. Our analysis will specifically focus on polysemy in terminology.

3. Structure of the resource

In this section, we present the structure of the resource. Specifically, in Section 3.1 we illustrate the general principles that underlie the structural development of the resource. In Section 3.2 we detail the parts that compose the resource.

3.1. *General principles*

The presented resource is a terminology resource. This implies that the structure is conceived as concept-oriented, instead of term-oriented. In particular, for the structure of the resource, we take as a reference the standard ISO 16642 (2017), which concerns the *Terminological Markup Framework* (TMF) meta-model. The application of the meta-model allows to obtain a hierarchical structure that consists of three different levels, conceived as follows: 1) the Terminological Entry (TE) refers to a concept; 2) the concept is represented in different languages in the respective Language Sections (LSs); 3) each Term Section (TS) specifically concerns the term in a determined working language and all information related to that term. The resource is structured following the same exact configuration, which consists of three levels: a conceptual level, a language level and a term level.

3.2. *Interaction with the resource*

The user groups (physicians, patients and language professionals) are presented with the same interface. Following the general principles, the interaction with the resource starts at the conceptual level. The user, indeed, is initially presented with a section dedicated to the conceptual dimension of terminology (see the circle numbered 1 in Figure 1, p. 31). In particular, the user has to first select a subject field and then select the “concept” to be analyzed. When the concept is selected, the list of the characteristics that make up the concept is shown in the subsection named “characteristics.”

The second section is related to the linguistic dimension of terminology (see the circle numbered 2 in Figure 1). First, the user is asked to select the working language, choosing between English, Italian, Spanish and French. For each language, the natural language definition of the concept and the reference link to the source are also comprised, respectively in the subsection “definition” and “source.” Then, the term(s) designating that concept in that specific

language are shown (see the circle numbered 3 in Figure 1). For each term, there is a subsection dedicated to the “signification.” In the resource, “signification” exclusively refers to the denotative sense of the analyzed term, that is to say its neutral meaning. The subsection shows the list of denotative semes in the form of a collection named “sememe.” Each denotative seme respectively corresponds at the semic level to one of the characteristics listed in the “characteristics” section. The user can then visualize the same types of information in a different working language, by additionally selecting another language.

Within the term subsection, there is also the additional information related to the set of connotative semes attributed to the analyzed term in the selected language. The sets of connotative semes, that differ from one another, are respectively indicated in different subsections named “connotation.” All the subsections “connotation” share the same set of denotative semes indicated at the “signification” level. Finally, a natural language explanation of the connotation attributed to the term is provided.

The resource additionally provides focused information concerning denotative and connotative semes. The user can specifically access this kind of information by clicking on each listed seme. Due to space limitations, a figure dedicated to the inventory of semes is not included in the article. The user can visualize the inventory of semes when interacting with the resource.

By clicking on a listed seme, a different terminological entry is displayed, that is entirely dedicated to the seme under analysis. The terminological entry contains three subsections, respectively named “subdomain,” “polarity” and “notes.” In the subsection “subdomain,” the medical subdomain in which the term – represented in the form of seme – is used is indicated. In the case in which a connotative seme is analyzed, the subsection “polarity” presents the indication concerning the polarity of the term, which can be either positive, negative or neuter. Finally, in the subsection “notes,” additional information related to the positive or negative connotation of the term is offered. In particular, an attestation or

motivation of the attribution of a positive or negative connotation to the term is provided, by resorting to quotations from specific cited works. In the following section, we conduct a case study in which the design of the resource is shown.

4. Case study

In this section, we present a case study aimed at illustrating the functionality of the resource. In particular, we aim at demonstrating the potentiality of the resource to systematically and distinctively represent the set of denotative semes and the set of connotative semes of medical terminology. At the same time, we also aim at highlighting the possibility offered by the resource to distinctively represent data concerning the conceptual dimension and the linguistic dimension of terminology. In Section 4.1, we conduct the semic analysis of the term “congenital heart defect,” by applying the methodology proposed by Bonato, Di Nunzio, and Vezzani (2024). The methodology, whose application is exemplified by analyzing “congenital heart defect,” underlies the development of the resource and can be therefore adopted to analyze all medical terms that possess connotation. In Section 4.2, we show the representation of the terminological data in the resource.

For the purposes of this study, we define the graphical signs used to respectively distinguish between the characteristics of the concepts, terms and semes. More specifically, the characteristics of the concepts are inserted between brackets, so that the characteristics are graphically represented in the following way: [characteristic of the concept]. Concepts are capitalized and inserted between single chevrons, as in <Concept>. Terms, instead, are written in lowercase and graphically represented by way of enclosing them between double quotation marks, as in the case of “term.” Each /seme/ is written in italics, and it is preceded and followed by the slash “/” character. When sememes are represented, the slash character that encloses the semes also performs the function of

separating each seme from the others. Figure 1 shows the design of the resource.

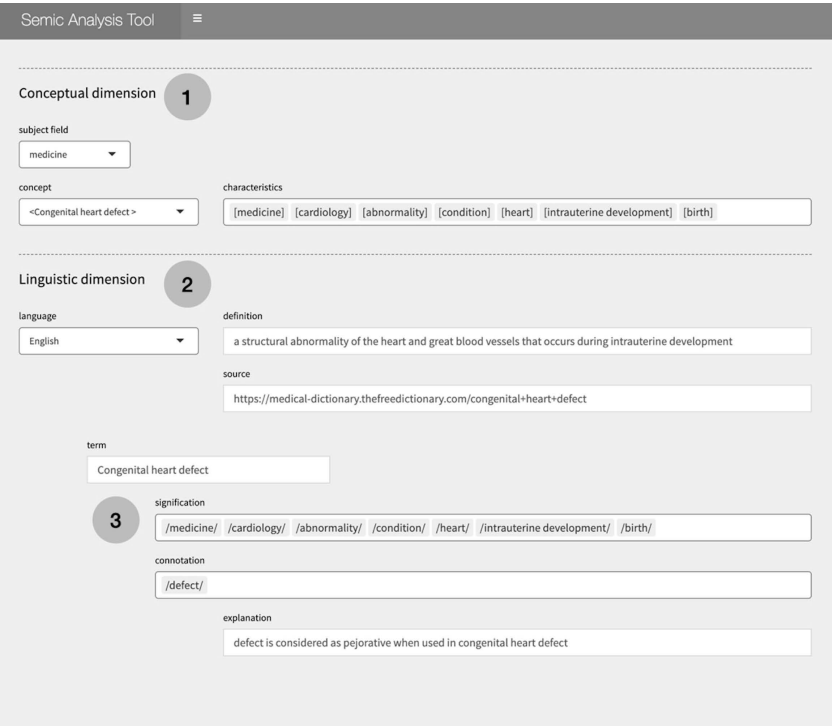


FIG. 1 – Screenshot of the interface of the Web application for semic analysis

4.1. *Semic analysis of “congenital heart defect”*

In this section, we present the semic analysis of the medical term “congenital heart defect” which, according to Dury (2012), possesses a negative connotation. To obtain the semic analysis of the term, we manually apply three of the four sequential steps that compose the methodology proposed by Bonato, Di Nunzio, and Vezzani (2024): 1) semi-automatic domain-specific corpus-based extraction of semes, 2) termhood to address the diaphasic and the

diastratic variations of language, and 3) automatic lemmatization of semes.

The application of the methodology starts with the recollection of the definitions of the concept <Congenital heart defect> from specialized dictionaries related to the medical domain. Specifically, we adopted English as the working language for the case study. For the purposes of the case study, we initially took as a reference two definitions that provide the denotation of the term. The first definition is taken from the *Medical Dictionary* contained in the *TheFreeDictionary's Medical Dictionary*:³

A structural abnormality of the heart and great blood vessels that occurs during intrauterine development. Abnormalities are commonly classified by the presence or absence of cyanosis. Acyanotic abnormalities include atrial and ventricular septal defects, coarctation of the aorta, and patent ductus arteriosus. Cyanotic defects include tetralogy of Fallot, transposition of the great vessels, and hypoplastic left-sided heart syndrome.

The second definition of the concept, instead, is contained in the *Dictionary of Medical Terms* (2010): “a heart condition existing at birth.” Taking as a basis these definitions, we reformulated them in order to create a single intensional definition that comprises the shared common elements. For the creation of the intensional definition, we particularly selected two generic concepts: <Abnormality> and <Condition>. Our choice is motivated by the consideration that <Condition> can be regarded as a superordinate concept of <Abnormality>. Then, we identified the delimiting characteristics of the concept, that are necessary for the comprehension and the representation of the concept at the conceptual level: [of the heart], [existing at birth] and [occurs during intrauterine development]. With reference to the characteristics [existing at birth] and [occurs during intrauterine development], we identify them as elements that should both be included in the intensional definition, notwithstanding their dissimilarity from a lexical, conceptual and linguistic

³ <https://medical-dictionary.thefreedictionary.com/congenital+heart+defect> (consulted 06/08/2024).

perspective. Indeed, we consider that the manifestation of an abnormality or condition at birth implicates that the abnormality or condition that affects the newborn originated in the course of the intrauterine development. As a matter of fact, as stated by Poly et al. (2018: 51), “[a] congenital heart defect is a heart problem which is present at birth, caused by improper development of the heart during fetal development.” By taking into consideration these elements, we obtained the following intensional definition, written in natural language: “abnormality or condition of the heart, which occurs during intrauterine development and exists at birth.”

Subsequently, we performed the extraction of medical terms from the intensional definition, to obtain the list of semes:

/medicine/ /cardiology/ /abnormality/ /condition/ /heart/ /intrauterine development/ /birth/

As can be observed, the characteristics that make up the concept at the conceptual level match at the linguistic level the denotative semes that compose the sememe of the term. We also added a seme that specifies the domain. Due to the fact that “congenital heart defect” is a medical term, we selected the seme */medicine/*. Moreover, we included in the sememe the seme */cardiology/*, that represents the medical subdomain in which the term is used. The analysis of the sememe highlights the exclusive presence of semes that express the denotation of the term “congenital heart defect.” Considering the approach proposed by Bonato, Di Nunzio, and Vezzani (2024), this sememe constitutes the “neutral meaning” of the term, which is a meaning that is not affected by any positive or negative connotation. This sememe, specifically, constitutes the signification of the term. In this work, we also analyze the connotation attributed to terms. We specifically consider terms as connotative semes when an attestation of the connotation attributed to terms is present in dictionaries or in scientific works.

In specific linguistic contexts of use, the term “congenital heart defect” acquires connotation. As indicated by Dury (2012), indeed, the term is perceived as pejoratively connotated by patients,

due to the fact that it brutally represents the designated concept. In particular, as stated by the author (*ibid.*: 89):

*L'utilisation de termes, qui à cause de leur connotation, portent un jugement quelconque sur le patient, ne peut guère correspondre à la position de surplomb et d'objectivité que l'on associe généralement à l'expertise.*⁴

For the purpose of avoiding the use on the part of physicians of the term “congenital heart defect,” the author proposes to rename the concept through the adoption of neological terms that do not carry a pejorative connotative judgment: “congenital heart anomaly” and “cardiovascular malformation.” As declared by the author (*ibid.*):

*[...] cette position d'expertise nécessite également que les termes utilisés donnent une idée juste et technique de ce qui est décrit : le terme cardiovascular malformation semble mieux convenir que celui de congenital heart defect, par exemple, parce qu'il exprime de façon plus juste et plus technique le fait que la cardiopathie congénitale est tout simplement une malformation de naissance.*⁵

More specifically, Dury (*ibid.*) highlights that:

defect [*est*] considéré comme péjoratif lorsqu'il est employé dans congenital heart defect [...].⁶

The term “defect” is present in the following three definitions of the concept <Congenital heart defect>: 1) “Developmental abnormalities involving structures of the heart. These defects are present at birth but may be discovered later in life,” taken from the

⁴ The use of terms which, because of their connotation, convey a judgement of any kind about the patient, can hardly correspond to the position of overhang and objectivity generally associated with expertise.

⁵ [...] this position of expertise also requires that the used terms give an accurate and technical idea of what is described: the term *cardiovascular malformation* seems more appropriate than *congenital heart defect*, for example, because it expresses more accurately and technically the fact that congenital cardiopathy is quite simply a birth malformation.

⁶ *defect* is considered as pejorative when used in *congenital heart defect* [...].

MeSH website;⁷ 2) “a structural defect of the heart or great vessels or both, present at birth,” contained in the *Miller-Keane Encyclopedia and Dictionary of Medicine, Nursing, and Allied Health* provided in the *TheFreeDictionary’s Medical Dictionary*;⁸ and 3) “a birth defect involving the heart,” extracted from the general language dictionary *Vocabulary.com Dictionary*.⁹

According to the definition provided by the *Farlex Partner Medical Dictionary* included in the *TheFreeDictionary’s Medical Dictionary*,¹⁰ the term “defect” designates “[a]n imperfection, malformation, dysfunction, or absence; an attribute of quality, in contrast with deficiency, which is an attribute of quantity.” The definition is preceded by the attestation of the fact that “[n]egative or pejorative connotations of this word may render it offensive in some contexts.” With respect to the term “defect” used to refer to birth conditions, Losorelli, Su-Velez, and Balakrishnan (2023) point out to the fact that it is preferable to make use of the terms “anomaly” or “variation,” considered as “less stigmatizing and equally clear, descriptive, and precise.” Furthermore, with specific reference to “birth defect,” both patients that are affected by it and their relatives “have voiced concerns that the word ‘defect’ might connote a negative perception and labeling” (Mai, Petersen, and Miller 2012: 984).

Taking these attestations into consideration, it seems necessary to formulate a sememe that also includes the seme /*defect*/. Indeed, “defect” conveys the negative connotation acquired by the term “congenital heart defect” in specific contexts of use. The following sememe, therefore, is composed by the same set of denotative semes and the connotative seme /*defect*/:

⁷ <https://meshb.nlm.nih.gov/record/ui?ui=D006330> (consulted 06/08/2024).

⁸ <https://medical-dictionary.thefreedictionary.com/congenital+heart+defect> (consulted 06/08/2024).

⁹ <https://www.vocabulary.com/dictionary/congenital%20heart%20defect> (consulted 06/08/2024).

¹⁰ <https://medical-dictionary.thefreedictionary.com/defect> (consulted 06/08/2024).

/medicine/ /cardiology/ /abnormality/ /condition/ /heart/ /intrauterine development/ /birth/ /defect/

This sememe, that differs from the first one due to the presence of connotation in it, constitutes therefore one sense that is contextually acquired by the term. In particular, the existence of two different sememes that are associated to the term “congenital heart defect” determines the polysemy of the term.

A further observation that stems from the analysis of the definitions of the term “congenital heart defect” is that the negatively charged term “defect” is included in a definition provided in the MeSH website, that constitutes a medical specialized resource. It can be therefore noticed that connotation attached to medical terms is not solely found in general language dictionaries. As a matter of fact, the ascription of connotation to medical terms is also attested in resources that reflect the definition attributed to a term from a medical specialized viewpoint. As Dury (2012) highlights, indeed, the “neological need” that arises because of the negative connotation attached to medical terms is not exclusively perceived by a single individual, as it concerns the collective conscience of a whole group of subjects belonging to the same scientific community.

4.2. *Data representation in the resource*

In this section, we illustrate the interface of the resource by offering a visualization of the representation of the data concerning the term “congenital heart defect.”

As can be seen in Figure 1, in the section named “Conceptual dimension” we selected the subject field medicine and the concept <Congenital heart defect>. The resource shows the characteristics that make up the concept in the subsection “characteristics.” The characteristics are [medicine], [cardiology], [abnormality], [condition], [heart], [intrauterine development] and [birth].

In the section “Linguistic dimension,” we opted for English as a working language by selecting “English” in the “language” sub-

section. This action implies that the resource displays the definition of the concept in English, provided in the subsection “definition.” In the subsection “source,” the indication of the link to the source from which the definition was collected is present.

It is then possible to visualize the term that designates the concept in English, that is “congenital heart defect.” The denotative sememe is shown in the “signification” subsection. The denotative semes are */medicine/*, */cardiology/*, */abnormality/*, */condition/*, */heart/*, */intrauterine development/* and */birth/*. As can be noticed, each characteristic correspond to one of the semes that compose the denotative sememe of the term.

Only one subsection named “connotation” is present. As a matter of fact, for what concerns the term “congenital heart defect,” only one sense that is attributed to the term was detected, expressed by the presence of the connotative seme */defect/*. The “explanation” subsection, instead, provides the attestation of the attribution of a negative connotation to the term “defect.” To give this information, we resorted to a quotation taken from the work by Dury (2012), in which an example of a terminological context in which the term “defect” assumes a negative connotative charge is provided.

Conclusion

In this work, we presented a multilingual terminology resource that focuses on semic analysis for medical terminology. The resource is conceived to distinctively represent both the denotation and the connotation of medical terms in the form of denotative and connotative semes, that are atoms of meaning. The respective thorough examination of both types of semes emerges as functional to acquire a sound knowledge of the semantic dimension of medical terminology. In particular, the resource provides the possibility to examine the potential positive or negative connotation attached to terms. The acquisition of this kind of information can be deemed useful on the part of physicians, patients and language professionals. Furthermore, the resource allows to systematically

observe the distinction between the conceptual dimension and the linguistic dimension of terminology. Indeed, information related to the analyzed concept and to the term that linguistically designates the concept are respectively included in different sections.

In the resource, we follow the approach proposed by Bonato, Di Nunzio, and Vezzani (2024) concerning the use of semic analysis to study polysemy and polyreferentiality. According to the authors, polysemous terms present the ascription of different sememes, of which one is exclusively denotative and the other(s) also contain connotative semes. Polyreferential terms, on the contrary, do not share the same concept, so that their sememes do not contain the same combination of semes. In particular, in the context of the article, we adopt this approach to analyze polysemy.

The resource also features the integration of an inventory of semes, aimed at providing additional information concerning the atoms of meaning that represent the denotation and the connotation of medical terms. By way of indicating the polarity of the terms, a more comprehensive knowledge of medical terminology can be achieved.

As future work, we propose to increase the quantity of available terminological data contained in the resource. This operation is essential to implement another functionality that can be provided by the resource, which is the automatic individuation of the semes that are shared between the sememes of multiple medical terms. In this way, it could be also possible to easily detect similarities or differences for what concerns the data related to the conceptual and linguistic dimension of medical terminology. Moreover, we aim at integrating additional working languages, to provide a more comprehensive analysis of the dual dimension of terminology and the connotative values ascribed to terms in different languages. The terminology resource, indeed, can potentially support medical terminological data related to all languages. Furthermore, we intend to collect user testing feedback from the three target user groups, to evaluate the efficiency and usability of the resource. We also

plan to conduct a user study to evaluate the visualization of data in the interface, and to optimize it on the basis of the obtained results.

Focusing on the inventory of semes, as future work, we propose to conduct an in-depth analysis of the denotation and connotation of medical terms. For instance, we propose to integrate a functionality that provides information concerning the connotation acquired by terms from a diachronic viewpoint. In particular, we aim at providing information concerning the first attestation of terms in specialized dictionaries and the first documented attestation of the acquisition of connotation on the part of the terms. In addition, we intend to adopt an automatic approach to the detection of medical term polarity by way of resorting to dedicated annotated datasets. Finally, taking as a basis the aforementioned statement according to which “[a]ll words can potentially convey affective meaning” (Strapparava and Mihalcea 2007: 70), we propose to investigate whether medical terms can potentially convey a neuter polarity.¹¹

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¹¹ The web application and the source code are available at <https://github.com/gmdn/SemicAnalysis> (consulted 06/08/2024).

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